2110322: Normalization Exercises

Exercise 1

Consider a table **R** with the following set of attributes and set of functional dependencies:

R(aircraftNo, aircraftType, capacity, customerNo, day, flightNo, seat)

customerNo, flightNo, day → seat
flightNo, day → aircraftNo
aircraftNo → aircraftType
aircraftType → capacity

Design a sequence of decompositions that transforms the table **R** into a set of tables that are all in BCNF.

Exercise 2

Consider the Students table, with the primary key underlined, and the following data:

Students:

ID	Name	Email	Courses	GradePoints
100111	John Doe	doe@usna.edu	NN204, SI204, IT221	2,3,3
092244	Matt Smith	smith@usna.edu	SM223, EE301	4,4
113221	Melinda Black	black@usna.edu	SI204	3
090112	Tom Johnson	Johnson@usna.edu	NN204, SI204, IT221	4,2,3

Identify the functional dependencies. Redesign the tables such that all the information currently in the Students table is found in the resulting tables, and the resulting tables are in BCNF.

Exercise 3

Below is the Rentals table created for the DVD-by-mail division of Neatflix.

RentalID	Title	CustomerID	MailedOutDate	Director	MovieCategory	Price
1	Die Hard	1001	3/3/2010	John McTiernan	Old	$4.25
1	The last man standing	1001	3/3/2010	Walter Hill	Old	$4.25
1	Wedding Crashers	1001	3/3/2010	David Dobkin	New	$5.50
2	Dodgeball	1002	3/4/2010	Rawson Marshall Thurber	New	$5.50
2	Die Hard	1002	3/4/2010	John McTiernan	Old	$4.25
3	As good as it gets	1003	1/7/2011	James Brooks	Old	$4.25
4	Forest Gump	1001	1/7/2011	Robert Zemeckis	Old	$4.25

The primary key of the Rentals table is the composite key (**RentalID, Title**). Also assume that the cost of all movies in a given category is the same.

Based on your general knowledge of movies and rentals, identify what you believe are reasonable functional dependencies for the Rentals table for a DVD-by-mail business. Then, decompose the Rentals table such that the resulting tables are in BCNF. For each of the resulting tables, give the table name, column names, primary keys.